SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

Item	Details
Course Name	Cloud Computing and Big Data Analytics
Course Code	CSA15
Course Outcome Covered	CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)
Assessment Tool Weightage	10%
Dave's Taxonomy	Manipulation (Level 2)
Question Count	5
Total Marks	25

Code of Conduct

I I Rupesh(192511170) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __Rupesh I__

Assessment Tasks: Application Based Questions

Question 1. *A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.*

Answer

1. Introduction

Seasonal sales such as Diwali, Black Friday or Big Billion Days can multiply traffic many times within minutes. A fixed on-premises server farm is either over-provisioned (wasted money) or under-provisioned (crashes at peak). Cloud elasticity solves this through auto-scaling, load balancing and continuous monitoring working together.

2. How Auto-Scaling Handles Spikes

- **Horizontal scaling:** An Auto Scaling Group (AWS), Virtual Machine Scale Set (Azure) or Managed Instance Group (GCP) adds or removes identical virtual machines or containers based on rules, for example "add 2 instances when average CPU exceeds 70% for 3 minutes".
- **Scheduled scaling:** For known events (a flash sale at 12 noon), capacity is pre-warmed before the event starts.
- **Predictive scaling:** Machine-learning models use past traffic patterns to launch capacity ahead of demand.
- **Scale-in:** When traffic falls, surplus instances are terminated, so the company pays only for what it used (pay-as-you-go).

3. Role of Load Balancing

A load balancer (AWS Elastic Load Balancer, Azure Load Balancer, Google Cloud Load Balancing) is the single entry point that distributes incoming requests across all healthy instances using algorithms such as round robin, least connections or weighted routing. It performs health checks and automatically stops sending traffic to failed servers. It can also terminate SSL, maintain session stickiness and spread instances across multiple Availability Zones so that no single data centre becomes a bottleneck.

4. Reducing Downtime and Improving User Experience

- No single point of failure: if one instance or zone fails, the load balancer re-routes users and auto-scaling replaces the failed capacity.

- Consistent response time: extra capacity is added before queues build up, so pages and checkout stay fast.
- Fewer cart abandonments and failed payments, which directly protects revenue and brand reputation.
- Zero-downtime deployments through rolling or blue-green updates behind the load balancer.

5. Role of Monitoring Tools

Monitoring tools such as Amazon CloudWatch, Azure Monitor, Google Cloud Monitoring, Prometheus and Grafana collect metrics (CPU, memory, request count, latency, error rate), logs and traces. They are the "eyes" of the system: metrics feed the auto-scaling policies, alarms notify the operations team through email or SMS, dashboards show live health during the sale, and historical data helps with capacity planning and cost optimisation.

6. Real-World E-Commerce Scenario

Consider an online fashion store running a midnight flash sale. Normal traffic of 2,000 users rises to 60,000 within ten minutes. CloudWatch detects CPU above 70% and request latency rising; the Auto Scaling Group grows from 4 to 40 web servers in a few minutes. An Application Load Balancer spreads traffic across three Availability Zones, while a read replica and a cache (Redis) protect the database. When one server crashes, health checks remove it within seconds and shoppers notice nothing. After the sale, the group scales back to 4 servers and the bill reflects only the hours actually used. Amazon and Flipkart rely on the same principles every sale season.

7. Conclusion

Auto-scaling provides elasticity, load balancing provides availability and fair distribution, and monitoring provides visibility and automation. Together they deliver a resilient, cost-efficient storefront with a smooth user experience even under unpredictable demand.

Question 2. *An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.*

Answer

1. Introduction

Backup data must survive hardware failure, human error, ransomware and site-wide disasters. Cloud object storage such as Amazon S3, Azure Blob Storage and Google Cloud Storage is designed for exactly these requirements, offering scalable, secure and low-cost long-term retention.

2. Durability and Data Redundancy

- **Replication:** Every object is automatically stored as multiple copies across several devices and Availability Zones. Amazon S3, for example, is designed for 99.999999999% (eleven nines) durability.
- **Erasure coding and checksums:** Data is split into fragments with parity information, and checksums continuously detect and repair silent corruption.
- **Geo-redundancy:** Cross-region replication (S3 CRR, Azure GRS) keeps a copy in a distant region to survive regional disasters.
- **Versioning and object lock:** Previous versions are retained, and write-once-read-many (WORM) locks prevent deletion or encryption by ransomware.
- **Lifecycle policies:** Data automatically moves from hot storage to cool and archive tiers as it ages.

3. Encryption and Access Control

- **Encryption in transit:** TLS/HTTPS protects data while it is uploaded or restored.
- **Encryption at rest:** AES-256 server-side encryption, with keys managed through a Key Management Service (AWS KMS, Azure Key Vault) or customer-managed keys. Client-side encryption lets the organization hold the keys exclusively.
- **Identity and access management (IAM):** Least-privilege roles and policies decide who can read, write or delete backups.
- **Multi-factor authentication and MFA-delete:** Add a second barrier before backups can be removed.
- **Network controls and auditing:** Private endpoints, bucket policies, blocking public access, and logs (CloudTrail) record every access attempt.

4. Cost Advantages over Traditional Storage

Aspect	Traditional (tape / SAN / NAS)	Cloud storage
Upfront cost	High capital expenditure for hardware and data-centre space	No upfront cost; operating expense
Scaling	Must forecast and buy capacity in advance	Virtually unlimited, pay for what is used
Maintenance	Staff, power, cooling, hardware refresh	Handled by the provider
Redundancy	Needs a second site, which doubles cost	Built in at low cost
Archive pricing	Tape handling and off-site vaulting	Very cheap archive tiers (e.g. S3 Glacier)

5. Disaster Recovery Use Case

A hospital network backs up patient records nightly to an encrypted bucket with versioning and cross-region replication. When ransomware encrypts the primary servers, the IT team uses an immutable, unaffected backup copy to restore data in the secondary region. They launch standby servers from pre-built machine images and redirect users using DNS failover. Recovery Point Objective (RPO) is under 24 hours and Recovery Time Objective (RTO) is a few hours, without buying an idle second data centre. Since access logs and encryption are in place, the hospital also meets its patient-data privacy obligations.

6. Conclusion

Cloud storage gives backups high durability through replication, strong confidentiality through encryption and IAM, and significant savings through pay-as-you-go pricing, making it the preferred foundation for modern disaster recovery.

Question 3. *A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.*

Answer

1. Introduction

When users in Mumbai, London and Sao Paulo all connect to one server in Virginia, distance adds delay (latency) and congestion. A Content Delivery Network (CDN) solves this by placing content and logic physically close to users.

2. How CDNs Improve Performance and Reduce Latency

- **Caching:** Static assets such as images, video segments, app updates, scripts and API responses are cached at hundreds of Points of Presence (PoPs). A request is answered from the nearest PoP instead of the origin.
- **Shorter network path:** Fewer hops mean lower round-trip time; intelligent routing (Anycast, DNS-based geo-routing) sends the user to the closest or best-performing node.
- **Origin offload:** Because most requests never reach the origin, bandwidth cost and server load drop sharply.
- **Optimisation:** Compression, HTTP/2 and HTTP/3, TLS termination and image resizing speed up delivery on mobile networks.
- **Security:** Distributed DDoS absorption and web application firewall at the edge.

3. Role of Edge Servers in User Experience

Edge servers sit inside or near ISP networks. They deliver faster app start-up and page loads, smoother video with less buffering, and quicker downloads of game patches. Edge computing (AWS Lambda@Edge/CloudFront Functions, Cloudflare Workers) also runs small pieces of logic, for example authentication checks, A/B testing, personalisation and localisation, close to the user. The result is consistent experience whether the user is in India, Europe or America, even on weak mobile connections.

4. How Cloud Providers Support Global Availability

- **Regions and Availability Zones:** Deployments can be replicated in multiple geographic regions, each with isolated data centres.

- **Global load balancing and DNS:** Services such as Amazon Route 53, Azure Traffic Manager and Google Global Load Balancer route users to the nearest healthy region and fail over automatically.
- **Multi-region data replication:** Globally distributed databases (DynamoDB Global Tables, Cosmos DB, Spanner) keep data close to users and available if a region fails.
- **Private global backbone:** Provider-owned fibre networks carry traffic more reliably than the public internet.
- **Compliance and SLAs:** Region selection supports data-residency requirements, and SLAs of 99.9% to 99.99% back availability.

5. Real-Time Streaming / Gaming Example

Take a global live-streaming and mobile gaming platform. Video is split into short segments and distributed through a CDN, so a viewer in Chennai pulls segments from a nearby Indian PoP instead of a far-away origin, giving fast start and adaptive bitrate without buffering. For multiplayer gaming, players are matched to the closest game-server region (for example Mumbai, Frankfurt, Virginia), keeping ping low; the CDN delivers patches and assets, while edge logic handles login and anti-cheat checks. If the Frankfurt region has an outage, global load balancing shifts players to another region. This is the approach used by Netflix, YouTube and online games such as Fortnite.

6. Conclusion

CDNs and edge servers cut latency and offload origin servers, while multi-region cloud infrastructure provides resilience and compliance. Together they let a mobile app company deliver a fast, reliable experience worldwide.

Question 4. *A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.*

Answer

1. Introduction

Banks and fintech firms handle highly sensitive data and are bound by regulations such as PCI DSS, SOC 2, ISO 27001, GDPR and, in India, RBI guidelines on data localisation and outsourcing. Cloud adoption is possible only when security, compliance and auditability are guaranteed.

2. Regulatory Compliance and Auditing Support

- **Shared responsibility model:** The provider secures the physical data centres, hardware and core services; the institution secures its data, identities, configurations and applications.
- **Certifications and attestations:** Providers hold independent audits (ISO 27001, SOC 1/2/3, PCI DSS, HIPAA) and publish them through portals such as AWS Artifact and Azure Compliance Manager.
- **Governance tools:** AWS Config, Azure Policy and Security Command Center continuously check resources against compliance rules and flag drift.
- **Audit trails:** Every API call and administrative action is recorded, giving auditors tamper-evident evidence.

3. Role of Logging, Monitoring and Encryption

- **Logging:** Services like AWS CloudTrail, Azure Activity Log and VPC flow logs record who did what and when, supporting forensic investigation and audits. Logs are stored in write-protected storage.
- **Monitoring and threat detection:** Tools such as CloudWatch, GuardDuty, Microsoft Defender and a SIEM raise real-time alerts on suspicious logins, unusual transfers or privilege escalation, enabling quick incident response and fraud detection.
- **Encryption:** Data is encrypted in transit (TLS 1.2+) and at rest (AES-256). Keys are held in a KMS or hardware security module (HSM), with rotation and, where required, customer-managed or bring-your-own keys.

- **Identity controls:** Least privilege, role-based access, MFA and segregation of duties limit insider and external risk.

4. Ensuring Data Sovereignty

- Choose cloud regions located inside the required country or jurisdiction, and restrict deployment to them using policies (for example region deny rules or Azure Policy allowed-locations).
- Disable cross-region replication and backup copies outside the permitted geography.
- Use customer-managed encryption keys stored in a local HSM so the provider and foreign authorities cannot read data without the institution's key.
- Review contractual terms, data-processing agreements and sovereign or dedicated cloud options, and keep audit evidence of data location.

5. Example: Sensitive Financial Transactions

Consider a bank processing UPI and card payments in the cloud. The payment API runs in an Indian region inside a private network; card data is tokenised and encrypted with AES-256 using keys from an HSM. All traffic is TLS-encrypted, and only roles with MFA can access the transaction database. Every access is logged to immutable storage, and an anomaly detection service raises an alert if a user suddenly pulls thousands of records or a transaction originates from an unusual location. Backups remain in a second Indian region to satisfy data localisation. During an RBI or PCI DSS audit, the bank produces the provider's certifications, its own access logs and policy reports as evidence of compliance.

6. Conclusion

Cloud providers supply the compliant infrastructure and tooling, but the institution must configure encryption, logging, access control and region policies correctly. Done properly, the cloud can meet and often exceed on-premises security and compliance standards.

Question 5. *A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.*

Answer

1. Introduction

Containerization packages an application with its libraries and dependencies into a lightweight, isolated unit that runs the same way everywhere. Tools such as Docker create containers, and Kubernetes manages them at scale.

2. Containers vs Virtual Machines

Feature	Virtual Machine	Container
Architecture	Each VM runs a full guest OS on a hypervisor	Containers share the host OS kernel; only the app and dependencies are packaged
Size	Gigabytes	Megabytes
Start-up time	Minutes	Seconds or milliseconds
Resource usage	High; each OS consumes CPU, RAM and disk	Low overhead; many more per host
Performance	Hypervisor and guest OS overhead	Near-native performance
Isolation	Strong, hardware-level	Process-level (namespaces, cgroups); slightly weaker
Best for	Different operating systems, strong isolation, legacy apps	Microservices, CI/CD, rapid scaling

In short, containers give higher density and faster scaling, whereas VMs give stronger isolation. Many organisations run containers inside VMs to combine both benefits.

3. How Orchestration (Kubernetes) Simplifies Deployment and Scaling

- **Automated scheduling:** Kubernetes places pods on suitable nodes based on available CPU and memory.
- **Auto-scaling:** The Horizontal Pod Autoscaler adds or removes pods based on load, and the Cluster Autoscaler adds nodes when needed.
- **Self-healing:** Failed containers are restarted, unhealthy pods are replaced, and workloads are rescheduled when a node fails.
- **Rolling updates and rollbacks:** New versions are deployed gradually with zero downtime and can be reverted instantly.
- **Service discovery and load balancing:** Services give pods stable names and distribute traffic automatically.
- **Declarative configuration:** YAML manifests describe the desired state; Kubernetes continuously enforces it. Secrets and ConfigMaps manage configuration safely.

4. Portability Across Cloud Environments

A container image contains everything required to run the application, so "it works on my machine" problems disappear. The same image runs on a developer laptop, on-premises servers and any cloud. Kubernetes is available as managed services (Amazon EKS, Azure AKS, Google GKE) and as open source, so manifests and Helm charts move between providers with little change. This reduces vendor lock-in, supports hybrid and multi-cloud strategies, simplifies migration and gives consistent dev, test and production environments.

5. Example: Microservices-Based Application

An online food-delivery application is split into independent services: User Service (login and profiles), Restaurant/Catalog Service, Order Service, Payment Service, Delivery Tracking Service and Notification Service. Each has its own container, database and release cycle, and communicates through REST APIs or a message queue, with an API gateway as the entry point. During dinner hours Kubernetes scales only the Order and Payment pods, while quiet services stay small. If the Notification Service crashes, it restarts without affecting orders, and the team can deploy a new Payment version with a rolling update. Netflix, Uber and Amazon follow similar architectures.

6. Conclusion

Containers offer lightweight, fast and portable deployment, and Kubernetes automates scaling, healing and releases. Together they are an ideal foundation for microservices and cloud-native development.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

How the Answers Meet the Rubric

Criteria	How each answer addresses it
Relevance to Application (25%)	Every answer begins from the stated use case (e-commerce sale, backup storage, global app, financial institution, development team) and ends with a scenario tied to it.
Use of Cloud Concepts (25%)	Correct use of auto-scaling, load balancers, durability and redundancy, CDN/edge, shared responsibility, KMS/HSM, data sovereignty, containers, Kubernetes and microservices.

Criteria	How each answer addresses it
Practical Insight (20%)	Real services (AWS, Azure, GCP, CloudWatch, S3, EKS/AKS/GKE) and real companies (Amazon, Flipkart, Netflix, Uber) are used in each worked example.
Completeness (15%)	Every sub-question in each task is answered under its own heading, including the final example the question asked for.
Clarity (15%)	Numbered headings, short bullets and comparison tables keep the explanation concise and easy to follow.